

PRACTICAL-08

CODE:

```
#include <stdio.h>
#include <stdlib.h>

int main()
{
    int i;

    // 1. malloc()
    int *arr = (int *)malloc(5 * sizeof(int));

    if (arr == NULL)
    {
        printf("malloc failed!\n");
        return 1;
    }

    printf("Memory allocated using malloc()\n");

    for (i = 0; i < 5; i++)
    {
        arr[i] = (i + 1) * 10;
    }

    printf("malloc array: ");
    for (i = 0; i < 5; i++)
    {
        printf("%d ", arr[i]);
    }
    printf("\n");

    // 2. calloc()
    int *zero_arr = (int *)calloc(5, sizeof(int));

    if (zero_arr == NULL)
    {
        printf("calloc failed!\n");
        free(arr);
        return 1;
    }

    printf("\nMemory allocated using calloc()\n");

    printf("calloc array: ");
    for (i = 0; i < 5; i++)
    {
        printf("%d ", zero_arr[i]);
    }
    printf("\n");
```

```

// 3. realloc()
arr = (int *)realloc(arr, 10 * sizeof(int));

if (arr == NULL)
{
    printf("realloc failed!\n");
    free(zero_arr);
    return 1;
}

printf("\nMemory resized using realloc()\n");

for (i = 5; i < 10; i++)
{
    arr[i] = (i + 1) * 10;
}

printf("After realloc: ");
for (i = 0; i < 10; i++)
{
    printf("%d ", arr[i]);
}
printf("\n");

// 4. free()
free(arr);
free(zero_arr);

printf("\nMemory released using free()\n");

return 0;
}

```

Screen shots:

```

sri_charan@SRICHARAN:~/2520090137_OSSP/2520090137_Practical$ gedit Practical_8.c
** (gedit:2570): WARNING **: 04:01:49.759: Could not load Peas repository: Typelib file for namespace 'Peas',
version '1.0' not found
** (gedit:2570): WARNING **: 04:01:49.759: Could not load PeasGtk repository: Typelib file for namespace
'PeasGtk', version '1.0' not found
sri_charan@SRICHARAN:~/2520090137_OSSP/2520090137_Practical$ gcc Practical_8.c -o Practical_8
sri_charan@SRICHARAN:~/2520090137_OSSP/2520090137_Practical$ ./Practical_8
Memory allocated using malloc()
malloc array: 10 20 30 40 50
Memory allocated using calloc()
calloc array: 0 0 0 0 0
Memory resized using realloc()
After realloc: 10 20 30 40 50 60 70 80 90 100
Memory released using free()
sri_charan@SRICHARAN:~/2520090137_OSSP/2520090137_Practical$ █

```